EV Market Performance & Insights

Project Report: Interactive Data Analytics & Strategic Dashboard with Tableau

Author: Naveen Patidar | **Date:** 28 October 2026 | **Domain:** CleanTech & Automotive Analytics | **Platform:** Tableau Desktop & Cloud

1.55B Units	$64,921	324.54 km	738K Tons
TOTAL GLOBAL SALES	AVG. VEHICLE PRICE	AVG. DRIVING RANGE	CO_2 OFFSET / YEAR

1. Introduction

1.1 Project Overview

The electric vehicle (EV) industry has emerged as one of the fastest-growing sectors in the global automotive landscape, propelled by strict emissions mandates, consumer demand for sustainability, and accelerated innovations in lithium-based electrochemical cells. This analytics initiative provides a robust, empirical examination of global market dynamics across 15 vehicle categories (including Hatchbacks, Sedans, SUVs, Two-Wheelers, and Heavy Electric Trucks) and key battery chemistries (LFP, NMC, NCA, Sodium-Ion, and Solid-State). By synthesizing operational, pricing, technical, and consumer satisfaction parameters, the project bridges the gap between raw automotive datasets and high-impact strategic intelligence.

The visual analytics architecture comprises multi-tier Tableau dashboards providing an executive cockpit for market performance, battery lifecycle mechanics, regional manufacturing geographies, and drivetrain decarbonization metrics.

1.2 Purpose & Business Value

The primary purpose is to convert millions of granular vehicle telemetry and sales records into actionable strategic pathways. The platform empowers manufacturers, strategy consultants, and clean-energy researchers to benchmark competitive positioning, identify margin-accretive vehicle categories, forecast range vs. charging efficiency trade-offs, and quantify verified emissions displacement worldwide.

2. Ideation & Problem Discovery

2.1 Problem Statement

Rapid diversification across modern EV vehicle segments and evolving battery chemistries has overwhelmed automotive decision-makers with fragmented data silos. Executive leadership frequently lacks a consolidated operational view integrating production volume, battery degradation life, driving range, and regional manufacturing shares. This leaves market strategists, management consultants, and environmental policy analysts operating without empirical cross-functional alignment.

2. Ideation Phase (Contd.)

2.2 Stakeholder Persona Analysis

Persona & Title	Organizational Role	Core Business Needs & Dashboard Expectations
Emily Johnson Market Strategy Director	Global EV OEM Manufacturer Strategy	Evaluates annual sales volume across vehicle segments, monitors category market share, analyzes customer ratings vs. range, and tracks battery chemistry adoption for portfolio planning.
David Chen Automotive BI Manager	International Automotive Advisory Firm	Monitors OEM market share, benchmarks competitive performance, tracks regional manufacturing footprints, and analyzes pricing tiers to advise new entrants on positioning.
Dr. Sarah Williams Mobility Research Analyst	Clean Energy Research Institution	Investigates battery degradation cycles, real-world driving range advances, global manufacturing concentration, and lifecycle CO_2 reduction to guide decarbonization policy.

2.3 Empathy Map Canvas

Dimension	Emily (Strategy Director)	David (BI Manager)	Dr. Sarah (Research Analyst)

Says	"Which vehicle categories justify capital allocation next quarter?"	"Who holds durable competitive advantages in regional distribution?"	"Which battery chemistry maximizes lifecycle sustainability and range?"
Thinks & Does	Focuses on margin optimization; monitors Hatchback and Sedan sales.	Validates OEM market claims; constructs data-driven market entry decks.	Correlates cycle degradation with range; publishes evidence-based models.
Feels	Pressure to defend market share against aggressive low-cost competitors.	Need for analytical precision and defensible advisory benchmarks.	Urgency to align technical EV metrics with global net-zero climate goals.

2.4 Solution Brainstorming & Scope Formulation

Exploratory synthesis isolated four foundational analytical tracks: *Commercial Velocity* (sales volume, revenue yield, brand rating), *Technical Specification* (range dispersion, charging power capacity, degradation % per cycle), *Geographic Production* (manufacturing cluster distribution across APAC, EMEA, and Americas), and *Decarbonization Impact* (cumulative CO_2 tons abated).

> **Core Architectural Takeaway:** The project consolidates these analytical threads into two responsive, linked Tableau dashboards paired with a guided Data Story, delivering tailored drill-downs for executive, consulting, and research personas.

3. Requirement Analysis & Data Engineering

3.1 Customer Journey Mapping

Journey Phase	User Action & Thought Process	Dashboard System Response
1. Awareness	Accesses executive portal to assess high-level market health.	Top KPI cards display Global Volume (1.55B), Avg Price ($64.9k), Range, and CO_2.
2. Segmentation	Filters performance by specific category (e.g. Hatchback vs SUV).	Dynamic cross-filtering adjusts production output, pricing, and volume bars.
3. Deep Dive	Examines battery longevity against degradation percentages.	Lifecycle scatter plots contrast cycle counts against degradation across chemistry.
4. Geo Analysis	Evaluates regional concentration and manufacturing origins.	Choropleth map highlights manufacturing density across India, China, Vietnam, etc.
5. Strategy Export	Translates insights into business cases or academic papers.	Interactive Tableau Public Story provides presentationready visual slides.

3.2 Solution Requirements

Functional Specifications: The system must deliver granular multi-attribute filtering across 15 vehicle classes; compute calculated fields for retained asset equity, battery degradation rates, and annual CO_2 offset formulas; and display coordinated tooltips with descriptive metadata.

Non-Functional Specifications: Sub-second filtering latency, color palettes optimized for visual accessibility, unified category legends, and full deployment compatibility with web-based Tableau Cloud / Tableau Public viewing nodes. **3.3 Technology Stack**

System Layer	Technology Utilized	Operational Function
Data Source	Cleaned EV Operational Dataset (.csv)	Normalized records covering sales, specs, OEM brands, and coordinates.
Data Modeling	Tableau Data Prep Engine	Type casting, outlier handling, calculated fields, and aggregations.
Analytics & Viz	Tableau Desktop Professional	Worksheet creation, visual encodings, LOD expressions, and dashboard integration.

Publishing	Tableau Public / Cloud Server	Secure cloud hosting and dynamic web stakeholder dissemination.

3.4 Data Flow Pipeline

Raw EV transactional logs → Schema parsing & calculated metrics (*Annual Sales Units*, *Retained Value*, *CO_2 Offset Sum*) → Dimensional worksheets (15 categories × 5 battery chemistries) → Interactive multi-view dashboards → Executive Data Story.

4. Visual Dashboard Implementation & Testing

4.1 Comprehensive Tableau Analytics Interface



Figure 1: Tableau Production Workspace showcasing category sales breakdown, manufacturer run rates, regional market share, battery tech range comparison, degradation lifecycles, and CO_2 offset.

4.2 Functional & Reliability Verification Testing

Verification Test Case	Design Specification & Baseline Expectation	Test Status	Verdict
KPI Aggregate Integrity	Sales = 1.55B; Avg Price = $64,921; Range = 324.54 km; CO_2 = 738K t.	Validated	**PASS**
Cross-Filter Synchrony	Category selection dynamically isolates dependent manufacturer bars and maps.	Validated	**PASS**
Lifecycle Deg. Scatter	Scatter coordinates faithfully plot cycle count vs. degradation % per battery chemistry.	Validated	**PASS**
Geo-Mapping Accuracy	Manufacturing facilities correctly geo-located across APAC, Europe, and Americas.	Validated	**PASS**
Web Rendering Response	All action filters and tooltips render under 1.2s on cloud publishing nodes.	Validated	**PASS**

5. Key Insights & Findings

Quantitative interrogation of the dataset via the interactive workspace yielded fundamental strategic benchmarks:

- **Volume Engines:** Hatchbacks and Sedans command high sales volumes among passenger vehicles, while TwoWheeler Scooters maintain continuous year-round transaction frequency.
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Battery Chemistry Trade-offs: Lithium Iron Phosphate (LFP) and NMC lead commercial adoption; however, SolidState chemistry demonstrates superior cycle lifespans (3,000+ cycles) and minimal degradation (<7.5%), representing the next high-value product horizon.

- **Manufacturing Concentration:** Production volume remains strongly anchored across Asian hubs (China, India, Vietnam), which serve as crucial export engines for international markets.
- **Asset Value Retention:** Commercial utility vehicles—specifically Electric Buses and Electric Trucks—demonstrate the highest 3-year retained capital equity ($14.5M and $8.69M respectively).

6. Strategic Evaluation (Pros & Cons)

Project Strengths & Business Advantages	Identified Constraints & Limitations
Unified cross-functional cockpit replacing disparate spreadsheets.	Analytical accuracy is constrained to static historical data batches.
Direct operational alignment for OEM, Strategy, and Academic personas.	15 granular categories create high chart density for mobile screens.
Transparent, cloud-hosted distribution via Tableau Public.	Does not capture localized public charging grid infrastructure.

7. Conclusion & Future Scope

This analytics project authored by **Naveen Patidar** demonstrates how robust business intelligence transforms complex automotive data into actionable strategic value. The interactive dashboard suite empowers executives like Emily Johnson to guide portfolio strategy, advisors like David Chen to benchmark OEM competition, and researchers like Dr. Sarah Williams to validate clean mobility impact.

Future Scope: Next iterations will integrate real-time API connectors for automated daily sales feeds, incorporate predictive machine learning models for battery degradation forecasting, and establish direct spatial layers for global DC fast-charging network saturation.

8. Project Appendix & Repository Deliverables

Deliverable	Repository & Live Reference Details
Author & Lead	**Naveen Patidar**
Tableau Public Demo	Interactive Dashboard Suite (Market Performance, Battery Life, Geo Footprint)
Source Dataset	Curated EV Global Fleet & Specifications Database (Normalized .CSV)
Version Control	Git Version Control Repository containing workbooks and schema definitions

11. Appendix

Source code(If any): Not applicable

Dataset link: used local.csv dataset

Project link: https://public.tableau.com/views/tab1_17930891578410/Dashboard1?:language=en-US&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link